Historical Failure Analysis Cases

Case 1: RM-001 Catastrophic Bearing Failure (2025-03-12)

Observation: Rolling mill seized during operation. Vibration sensors indicated 7.2 mm/s shortly before failure. Root Cause: Lack of lubrication due to blocked grease line. Resolution: Replaced bearing and installed automated grease flow monitors. Recommendation: Inspect grease lines monthly.

Case 2: BF-002 Tuyere Burnout (2024-11-05)

Observation: Water leak into the furnace causing a steam explosion. Root Cause: Tuyere cooling water flow was restricted due to scale buildup. Resolution: Replaced tuyere and chemically cleaned the cooling circuit. Recommendation: Implement daily cooling water flow checks and annual chemical cleaning.

Case 3: CV-001 Conveyor Belt Tear (2025-01-20)

Observation: Main feed conveyor belt tore laterally. Root Cause: A sharp piece of scrap metal bypassed the magnetic separator and jammed against the idler. Resolution: Spliced the belt and replaced the magnetic separator with a stronger unit. Recommendation: Regular calibration of the magnetic separator.